

Academic integrity considerations of using ChatGPT in assessment activities among university student teachers

Quality
Assurance in
Education

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Received 1 June 2024
Revised 24 August 2024
23 November 2024
1 December 2024
Accepted 2 December 2024

Abstract

Purpose – This study aims to investigate academic integrity considerations of Generative Pre-Trained Transformer (ChatGPT) usage in assessment activities among the finalist student teachers at the University of Dar es Salaam. The study builds upon other previous studies on the topic of artificial intelligence (AI) tools. However, it is unique in terms of its context and the focus on integrity.

Design/methodology/approach – A questionnaire was used to collect data from 383 finalist student teachers, and other 30 students were involved in focus group discussions (FGDs). Data for this study were analysed using IBM-SPSS Statistics Versions 25 to obtain descriptive statistics for the quantitative part and thematic analysis was used along with QDA Miner Lite for the qualitative part to generate themes.

Findings – Student teachers consider ChatGPT as a transformative AI technology to ease the accomplishment of their assessment tasks. However, a large percentage of them did not pay much attention to academic integrity while using the AI tool. About 199 (52%) out of 383 respondents said ChatGPT is a useful tool in generating answers for their assessment tasks compared to 192 (50.1%) of respondents who had negative opinions.

Research limitations/implications – Given the sensitivity of the topic on academic integrity some participants were hesitant to share information until when the researchers clarified the purpose of the study. Participants were told that the information they would provide was purely for academic purpose. Also, this study was conducted when students' regular classes were in progress, such that it was challenging to set universal time to conduct FGDs where all intended participants could attend. The researcher was flexible enough to find hours which were conducive for participants to participate in FGDs.

Practical implications – For practitioners, given the fact that ChatGPT is a relatively new AI tool, training for raising awareness of its use should be conducted. Besides, specific academic integrity guidelines for its ethical use should be formulated and implemented. Likewise, instructors should set questions that prompt students original thinking which the AI tool cannot easily give contextualized solutions.

Social implications – The university where the AI tool is used is an open system that is within the society, and the impacts of AI technologies are inevitable in social domains. The society, therefore, should be informed about the emerging AI tools with their associated potentials and shortcomings to safeguard the norms and interests of the society.



Quality Assurance in Education
Vol. 33 No. 2, 2025
pp. 305-320
© Emerald Publishing Limited
0968-4883
DOI 10.1108/QAE-06-2024-0100

Funding: This work was supported by the Mwalimu Nyerere Memorial Academy who sponsors my PhD studies.

Originality/value – The study provides a comprehensive overview of student teachers’ ChatGPT use and its implication on academic integrity. Given the novelty nature of ChatGPT use in education, course instructors and students may benefit from insights into responsible utilization of this AI technology without compromising integrity and quality of education.

Keywords Academic integrity, ChatGPT, Student teachers, Assessment activities

Paper type Research paper

1. Introduction

The application of artificial intelligence (AI) has captured the attention of many stakeholders in the sector of education. Massive growth and use of AI such as Generative Pre-Trained Transformer (ChatGPT) are greatly altering educational practices including the area of assessment. The primary aim of using technology in education is to improve the quality of education. Quality education is vital for a country in improving the quality of its human resources. Thus, the role of technology is pivotal in education because it simplifies the process of teaching and learning. However, care should be taken because any technology is a double-edged sword that could be constructive or destructive. Generally, the use of AI technologies in education should not compromise the principle of academic integrity. Adhering to academic integrity is imperative to have meaningful assessments that draw on measuring the level of students’ knowledge and realisation of learning outcomes. Assessment tasks are inherently embedded in and guided by the principle of academic integrity. In this regard, the irresponsible use of ChatGPT raises ethical concerns that underlie academic integrity. Among other AI tools, ChatGPT was chosen in this study because it was popular among student teachers and the most used due to its accessibility, ease of use, user-friendliness and accuracy in responding to queries. The generated responses could be in the form of multiple languages including English, German, Spanish and French (Deng and Lin, 2023). With this regard, it was unknown about students’ awareness of academic integrity considerations while using ChatGPT. Hence, the current study aimed to investigate the perceptions of academic integrity considerations of ChatGPT usage in assessment activities among finalist student teachers at the University of Dar es Salaam. The study was guided by the following two major questions:

- Q1. To what extent is ChatGPT useful for completing assessment tasks among the finalist student teachers?
- Q2. To what extent are finalist student teachers aware that the irresponsible use of ChatGPT in completing assessment tasks is against academic integrity?

2. Literature review and definitions of key terms

2.1 Academic integrity

The term integrity originated from the Latin adjective *integer*, referring to the state of being whole or complete (Monga*, 2016). Integrity is portrayed as the quality of one’s action to follow the existing norms, rules and moral values of a particular domain. It requires one to work in line with the tenets such as accountability, honesty, justice, impartiality and incorruptibility (Huberts, 2018). In particular, academic integrity is the state of “upholding honesty, trustfulness and strong moral principles by all members of the university community (students and both academic and administrative staff) in all academic endeavours” (TCU, 2019, p. 8). It is the state of maintaining rules and regulations that

describe the degree to which an individual is ethical and honestly upright while working on academic matters.

2.2 ChatGPT

ChatGPT is the form of AI. The notion of intelligence is initially associated with the human mental capability to understand, learn and apply knowledge to practice. Thus, the term is portrayed as one's ability to work effectively by using their understanding (Kaliraj and Devi, 2021). Also, from a psychological point of view: Gardner (1983, p. 12), for example, defines intelligence as "the ability to solve problems or to create products, that are valued within one or more cultural settings". Besides, intelligence refers to the "computational part of the ability to achieve goals in the world. Varying kinds and degrees of intelligence occur in people, animals and some machines" (McCarthy, 2004, p. 2). The latter definition suggests that the concept of intelligence has evolved to encompass not only humans and other animals but also machines as part of AI. Therefore, AI is a human-made ability of the mind to operate similarly to a human being enabled by the work of computer technology to comprehend meanings and acquire knowledge and its application (Lim *et al.*, 2023). Alternatively, AI is the reproduction of human cognitive abilities through the application of science and technology (Holmes *et al.*, 2022). This is contrary to human natural intelligence. AI can perform various tasks in different disciplines ranging from simple to complex problem-solving abilities. ChatGPT is an AI tool launched in November 2022. It is a chatbot to which people drop their queries and receive answers (feedback) instantly (Panagopoulou *et al.*, 2023). The tool has evolved from GPT-1 in 2018 to GPT-4 in 2023 as a language model that allows people to interact with a computer more naturally and conversationally on various topics (Sarrion, 2023).

2.3 Assessment activities

These are a series of related activities designed with several criteria intended to measure students' achievement of learning objectives of a particular subject matter (Martínez *et al.*, 2012). Assessment tasks need to be effective and efficient to determine the extent to which students have achieved the intended learning objectives for a given unit of learning. They are useful for making evaluations of students' knowledge, skills, experiences and competencies to establish the trustworthiness and relevance of university education. Thus, for the assessment activities to yield constructive and genuine feedback, they need to be guided by the principle of academic integrity.

2.4 The history of artificial intelligence

AI has existed for many decades ago, however, its modern era can be traced back to the 1950s (Deng and Lin, 2023). AI was invented by John McCarthy, a computer science and emeritus professor at Stanford University. According to him, AI is the science and engineering of making intelligent machines particularly intelligent software programs (Kaliraj and Devi, 2021). It is the branch of computer science that uses computers to study human intelligence. In the 1940s the mathematician Alan Turing introduced an early intelligence-based method of computing. He made a test on AI research with a notable question "Can machines think?". In 1956, John McCarthy triggered the commencement of a new epoch of research studies on AI through his definitions and views. His definitions made a concrete foundation for AI application. In its evolution, AI has borrowed numerous concepts, views and experiences from a variety of disciplines including computer science, mathematics, psychology, philosophy and linguistics. For instance, from computer science, AI has inherited programming techniques; the human mind and behaviour from psychology,

logical reasoning and computation from mathematics and from philosophy AI inherited common sense and mind (Kaliraj and Devi, 2021; Yagawa and Oishi, 2021).

Since its existence, AI has evolved and it experienced various notable landmarks. For instance, its application in business securities, cyber-attacks and government security systems are some of its achievements (Kaplan, 2024). Besides, AI has gained popularity through computerized programs for playing games and television quiz shows in the past two decades. Another well-known milestone of AI was in 2022 whereby AlphaGo was created by researchers at Google. They developed a program known as AlphaFold which was able to predict millions of proteins recognized in science. Before this invention, expensive devices and more years were spent for each protein to be analysed. In November 2022 OpenAI company built a large-scale application of AI and launched ChatGPT (Kaplan, 2024). ChatGPT was trained on a large collection of English-language texts from the internet which enabled it to engage in natural conversation.

2.5 Theoretical framework

This study was guided by constructivist theory. During the 1950s when computer application in education started to grow, the behaviourism learning theory by Skinner dominated classroom instructions. Behaviourism advocates learning as conditioning, whereby it is possible to explain human behaviour in terms of responses to stimuli (Yelland *et al.*, 2008). Consequently, this theory became linked with the idea of the computer machine as a teaching tool which gave rise to computer assisted instruction (CAI) (Romeo, 2006). The use of computers for instructing learners was expected to stimulate and motivate them to receive information which eventually led to the changes of behaviour. During the 1970s cognitivism rooted in cognitive psychology challenged Skinner's behaviourist approach to learning (McNeil, 2007). Under cognitivism a foundation for the thinking and understanding that learning is a construction of knowledge was manifested. During this period, the innovation of the first computer known as the IBM 1500 developed purposely for CAI took place (Romeo, 2006).

The 1980s witnessed the birth of the Apple 1 computer which went on sophisticating CAI. Thus, by the end of the 1980s, the dominant view on the use of computers in education was no longer limited to assisting instruction only but to be used by learners for inquiry, analysis and problem-solving (Yelland *et al.*, 2008). Under this view, a learner-centred approach was in place as per the cognitivist understanding that an individual can independently construct knowledge. The 1990s was the era of constructivism as an extension of cognitivism which emphasized inner mental constructions. During this era, the birth of the World Wide Web (WWW) and the drastic flow of digital multimedia favoured constructivist views. This is to say, a rich environment, surrounded by a variety of objects is an important factor for learners to learn and develop new knowledge. As such, the advent of WWW, the advancement of multimedia, networked technologies and digital communication tools opened opportunities for online learning and PowerPoint presentation (Schunk, 2004).

In line with constructivist dogma the focus was on knowledge construction and not just the consumption of digital artefacts. Recently, the development and use of computing systems, AI technologies, networking and telecommunication are ubiquitous. AI technologies are useful for supporting personalized student learning to enable them to construct knowledge. Thus, the use of AI in various educational processes resonates with constructivist learning theory, emphasizing the importance of students' knowledge construction and active engagement in learning (Blikstein and Worsley, 2016). In its broader sense, AI technologies including ChatGPT can support active learning through academic problem-solving, independent learning and students' interaction. With AI students can learn at their own pace while interacting actively with AI tutors.

2.6 ChatGPT use in university education

Numerous studies have proven that ChatGPT is useful in various teaching and learning activities (Green *et al.*, 2022; Halaweh, 2023; Hamal *et al.*, 2022; Lim *et al.*, 2023). ChatGPT can be used in the following ways: to facilitate educational planning and administration, generate personalized learning resources, provide instant feedback to instructors and learners about various educational problems, generate educational content, especially assessment tasks and help in grading them (Foltynek *et al.*, 2023; Shafik, 2023). Further, it facilitates communication and language learning, online tutoring to students and code generation for various data (Ray, 2023). The tool can be couched to understand the requirements of a paper or assignment and it is capable of providing suggestions and corrections for improvement of the academic work. This suggests that the tool is good at the data which is trained on, in contrast, it can be limited to comprehend or provide answers to some types of prompts or may provide biased answers. Therefore, to help the AI tool provide the most relevant output the user is required to provide context and use unbiased, clear and specific language when conversing with the ChatGPT (Zhai, 2022).

2.7 The use of ChatGPT by university students

ChatGPT can assist a student in writing an essay quickly or obtaining answers for assignments (Vaccino-Salvadore, 2023). This is because, ChatGPT is capable of writing essays, providing answers, writing stories, summarising information and comprehending specific topics (Farhi *et al.*, 2023; Su and Yang, 2023). Additionally, it improves communication skills, especially writing skills through a range of practical skills and it provides language translation instantly (Farhi *et al.*, 2023; Shafik, 2023).

The ability of ChatGPT to provide answers was also tested by the researcher in this study. The researcher asked two questions to ChatGPT as follows: Question one: *When did academic cheating begin?* and Question two; *What justifies academic cheating to be the right action?* The AI tool responded to these two questions quickly in less than 5 min. The first question was answered in 4 min, while the second one took only 1 min. The relevance and time of responses can vary due to the complexity and specificity of the query.

In a similar vein, Zhai (2022) conducted a study to find out the ability of ChatGPT to generate an academic paper on “artificial intelligence for education”. The results indicated that the AI tool was able to complete the work in two or three hours with limited effort from the author’s expertise. Some of the key uses of ChatGPT in education that were mentioned in the paper include grading and assessment of student tasks, student record management and provision of individual instructions to students. The foregoing practical experiences of ChatGPT usage substantiate its quick ability to generate ideas and respond to queries. The researchers’ conversation with ChatGPT in the current study is presented in Table 1.

Despite its benefits, unethical and irresponsible use of ChatGPT undermines the assessment process. It also affects the realization of the major purpose of university education of generating knowledge among its students. ChatGPT has been used to complete written coursework on behalf of students hence raising the question about academic integrity. Students who complete assignments using ChatGPT without citing, referencing and putting their critical voice to the information, gain an unfair advantage over their colleagues.

Students do not bother to admit the use of ChatGPT, they pretend that the work is theirs and consequently, they cheat (Perkins, 2023; Ipek *et al.*, 2023). The work generated directly from ChatGPT lacks originality, citations and references, and sometimes possesses incorrect responses (İpek *et al.*, 2023; Mhlanga, 2023; Sallam, 2023). Overreliance of ChatGPT endangers creativity and critical thinking among students because it retrieves information quickly from a multiple data set which eases the work of the student (Li *et al.*, 2023;

Table 1. ChatGPT Conversation with the researcher

Date and time of enquiry	Researcher’s question	Chat GPT response	Date and time of response	Response waiting time
[15 / 03/2023, 12:32]	When did academic cheating begin?	Academic cheating, or the act of using unauthorized or dishonest means to Fulfil academic requirements, has likely existed for as long as formal education has been in existence. There are records of students cheating on exams as far back as ancient Greece and Rome. For example, roman philosopher Seneca the younger described cheating as a common practice among students in his writings. However, the problem of academic cheating has become more pronounced in modern times due to factors such as increased competition for admission to colleges and universities, the pressure to succeed, and the availability of technology that can facilitate cheating	[15/03/2023,12:36]	4 min
[15/03/2023, 20:39]	What justifies academic cheating to be a right action?	Academic cheating is never a justifiable or right action. Cheating undermines the fundamental principles of education, which are based on trust, honesty and fairness. It also devalues the accomplishments of those who have worked hard to earn their grades and undermine the integrity of academic institutions. In addition, cheating can have serious consequences, such as academic penalties, loss of scholarships and awards, expulsion and damage to future academic and professional opportunities. Rather than resorting to cheating, students should seek help if they are struggling with coursework and should work hard to develop the skills and knowledge needed to succeed academically	[15/03/2023, 20:40]	1 min

Source: ChatGPT output

Mhlanga, 2023; Vaccino-Salvadore, 2023). There is a challenge in maintaining the quality of the feedback because ChatGPT and other language models are not faultless, they may produce errors. Therefore, tasks that require a high level of precision become limited. Improper use of ChatGPT interferes with the goal of assessment, that is to obtain genuine feedback about an individual’s mastery of knowledge (Holmes *et al.*, 2022).

Given these fast-moving innovations in AI, universities face the challenge of maintaining academic integrity to achieve quality education which is mostly exhibited through proper administration of assessment activities. Universities are responsible for providing high-quality education to meet the various needs of the local and international labour market.

This is in line with the United Nations' 2025 sustainable development goal (SDG) – 4 of “quality education” The major goal of SDG-4 is to ensure that everyone, regardless of their race, gender, age or other characteristics has access to inclusive and quality education (Küfeoğlu, 2022, p. 258). To achieve this responsibility, higher education including universities are regarded as dependable entities for the realization of sustainable development through the provision of relevant and quality education.

2.8 Previous studies on perceptions of ChatGPT use

A quantitative study was conducted in Croatia to investigate the following: the prevalence of ChatGPT usage among university students for written assessment tasks, ways in which the students use the AI tool, and students' views on the ethical consideration of its use. The study used a questionnaire to gather data via an online platform whereby 201 university students were involved. The study revealed that 55.2% have used ChatGPT in various ways, 81.9% used it for generating ideas, 18% used ChatGPT to produce complete assignments and for the case of ethical use of the tool, 85.2% were in favour of using it for idea generation (Črček and Patekar, 2023). The generation of ideas through ChatGPT occupied a huge percentage of university students in Croatia. This suggests that students do not depend exclusively on the AI tool to complete their academic tasks. This study has shed light on the current study by adding insights on ChatGPT use. However, it lacked the component of qualitative inquiry in which in-depth responses about the respondents' decision to use or not to use the AI tool for various educational needs could have been revealed.

Another quantitative approach and peer interview study explored students' experiences, perceptions and understanding of ChatGPT in China. The study involved 106 undergraduate students and 81 postgraduate students. The results indicate that 70.37% of postgraduate students portrayed a negative ChatGPT usage while 8.64% of undergraduate students held a positive view that ChatGPT is reliable (Xu *et al.*, 2024). This variation might be caused by academic maturity and experience between the two levels of education. The authors believe that postgraduate students have a strong orientation on matters pertinent to academic integrity due to their academic maturity and long experience in their academic trek. Postgraduate students might be in a good position to assess the strengths and weaknesses of ChatGPT, unlike their undergraduate counterparts.

Since its release, ChatGPT has attracted numerous authors and scholars in academia to investigate into it. The major focus of previous studies has been perceptions about the use of ChatGPT in education, potential benefits in teaching and learning and threats to academic integrity. However, given the fact that the use of ChatGPT is relatively a new phenomenon in higher education, there is still a significant void, especially in academic integrity considerations of student teachers in using ChatGPT in assessment activities.

2.9 Ethical and responsible use of ChatGPT

Consideration of the ethical and responsible use of ChatGPT is significant to obtaining authentic assessment results. Students need to know that ChatGPT is an artificial intelligent model with the ability to generate text to respond to various queries, and the end product from it needs to be critically examined (Li *et al.*, 2023). Students need to know that content from ChatGPT should be used for idea generation rather than treating it as a principal standalone end product (Hasanein and Sobaih, 2023). Students should be trained on both potential and threats when using chatGPT. Being aware of the side of threats, they will be in a position to judge what is accurate and useful for them as far as academic integrity is concerned. Similarly, instructors need knowledge about the use of this AI technology in assessment tasks so that they can be able to recognize the works generated by ChatGPT and grade them fairly (Lo, 2023). Instructors should state clearly in the course outline or on

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assessment tasks that the use of ChatGPT should be responsible and ethical. To that effect, there is a need for assessing its weaknesses in terms of inaccurate, biased and incomplete information they might produce at the same time reflecting on principles of academic integrity (Nam and Bai, 2023).

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3. Methodology

This study used a convergent research design under a mixed methods approach. The advantage of using this approach is that the weaknesses of the quantitative approach were supplemented by the strength of the qualitative (Bryman, 2012). In this study, the researcher gathered both qualitative and quantitative data in a single phase. Data were analysed separately and merged to find out whether they corroborated each other. Research tools were examined by experts in the topical area studied to see whether they adequately represent the content in appropriate proportions. Recommendations for changes to the emerging errors and ambiguities were made. Further, the researcher piloted the tools to a group of individuals who were similar in attributes to those who participated in the actual study. After the pilot study, the researcher translated the questions from English to Kiswahili and back to English to establish reliability.

The study used a sample of 383 finalist student teachers, 151 (39.4%) male and 232 (60.6%) female in the quantitative part who were systematically selected. The cohort of respondents was enrolled in the 2021 / 2022 academic year. By the time when ChatGPT was launched in November 2022, they were in the first semester of their second academic year. The finalist student teachers were chosen because they were upcoming professionals and agents of moral values in the field of education at work place. Also, it was fruitful to use finalist student teachers because they had experienced various incidents against academic integrity, including irresponsible use of ChatGPT. A questionnaire comprised of closed-ended questions was used. Also, focus group discussions (FGDs) were conducted with 30 participants who were part of the respondents who completed the questionnaire. They were purposively and conveniently selected to participate in the FGD. For the case of purposive selection, the researcher ensured the inclusion of participants who had used ChatGPT to complete their assessment tasks. Also, the convenient part of the selection considered the student teachers who voluntarily showed interest in participating in FGDs.

Quantitative data were analysed using IBM-SPSS Statistics Versions 25 to obtain descriptive statistics such as frequencies, percentages and cumulative percentages. Qualitative data obtained from FGD were analysed using thematic analysis with the help of QDA Miner Lite qualitative data analysis software in which three major steps were followed in the analysis including code generation, categorizations of codes and theme formation.

The distribution of respondents according to their programmes is indicated in Table 2.

4. Findings

4.1 The use of ChatGPT for completing assessment tasks

Student teachers are familiar with ChatGPT and they use it in various assessment activities. In this regard, the results show that student teachers seem to have both positive and negative perceptions towards the use of ChatGPT in the accomplishment of their assessment tasks. In the quantitative part of the enquiry, respondents were asked about their opinions on the use of ChatGPT to complete assessment tasks. The majority of student teachers, 199 (52%) have a positive view that ChatGPT is useful in completing assessment tasks and 184 (48%) seem to differ. The results of student teachers' views are depicted in Table 3.

For the qualitative part, participants were asked to give their views about the use of ChatGPT in completing assessment tasks. Again there were mixed feelings among student teachers about the use of ChatGPT to complete assessment tasks. Some participants had a

Table 2. Distribution of respondents per degree programme

Programmes	Counts (N = 383)	%	Valid percent	Cumulative percent
Bachelor of arts with education	270	70.5	70.5	70.8
Bachelor of education in adult and community education	11	2.9	2.9	73.6
Bachelor of education in commerce	3	0.8	0.8	74.4
Bachelor of education in early childhood education	21	5.5	5.5	79.9
Bachelor of education in physical education and sports sciences	20	5.0	5.0	84.9
Bachelor of education in psychology	13	3.4	3.4	88.3
Bachelor of science in education	45	11.7	11.7	100.0
Total	383	100.0	100.0	

Source: Authors' own work

Table 3. Perceived usefulness of ChatGPT for completing assessment tasks

Responses	Counts	%	Valid percent	Cumulative percent
Yes	199	52.0	52.0	52.0
No	184	48.0	48.0	100.0
Total	383	100.0	100.0	

Source: Authors' own work

positive perception of the use of ChatGPT in accomplishing their assessment tasks. They perceive the advent of this AI tool as a golden opportunity for them to easily process their assignments. For them, AI technology is a blessing since they harness it to solve their academic problems at any time of their need. Participants appreciated ChatGPT as a notable time-saving tool due to its instant nature of responding to their queries. Other participants appraised the AI tool as being user-friendly and not putting one under pressure since almost every question submitted to the AI tool is attended to accordingly.

The extracts below present some participants' comments on the positive views about the honest use of ChatGPT to complete their assessment tasks:

"We believe that work generated from ChatGPT can go without any doubt because it is wellcrafted to address different questions. This is a wonderful tool because it has answers for almost every question" (Response from ST-FGD, December 16th, 2023).

Other Participants in FGD had this to say:

"We do not waste much of our time to find answers, ChatGPT completes it on our behalf. Thus, using it is not cheating, instead, it is a tool like other tools such as 'Ask Me' and 'AIChat' which provide answers. Although instructors discourage copying and pasting from ChatGPT and other AI tools, still we find ChatGPT useful, user-friendly and flexible. Only that you need the internet connection to be able to obtain answers" (Response from ST-FGD, January 2nd, 2024).

In the foregoing extracts, participants admit the usefulness of ChatGPT in the accomplishment of their assessment activities. With reference to both approaches, there was a high corroboration between qualitative and quantitative data.

4.2 ChatGPT usage and academic integrity

Student teachers were asked about their perceptions of whether using ChatGPT to complete assessment tasks is against academic integrity. Findings indicated that 192 (50.1%) of student teachers who completed the questionnaire did not consider ChatGPT use as academic cheating, 174 (45.4%) perceived it as academic cheating, while 15 (3.9%) did not know. The students’ views are indicated in Table 4.

In qualitative data, some participants in FGDs had the opinion that the use of AI tools is not academic cheating because it provides relevant information for their academic activities. On the other hand, the majority of participants considered the use of ChatGPT as an enemy of academic integrity because it enabled them to present ideas which were not part of their original thoughts. They depended solely on ChatGPT to work on their behalf. Thus, partly qualitative data did not corroborate with quantitative data. Below are some of the participants’ negative thoughts about the use of ChatGPT:

“We don’t like those technological software including ChatGPT because not always they can provide the right answers and also they kill someone’s natural cognitive ability. Use of ChatGPT hinders the acquisition of required qualifications among student teachers when they leave the university. Also, failure of ChatGPT to provide citations and references reduces its credibility for being a dependable source of academic information” (Response from ST-FGD, 14th December 2023).

Further, participants added that:

“One day, students’ turn-up to the lecture was very low, the course instructor decided to provide a 10-mark quiz. Since we were unprepared, we decided to find answers from ChatGPT software. Everyone copied and pasted the answers from the tool. When the instructor marked the work he awarded 0 marks to every group member, except the one who paraphrased from the original text scored some marks. Overdependence on ChatGPT perpetuates illiteracy because the student does not utilize his or her abilities. Eventually, in future career the students become incompetent in their area of specialization” (Response from ST-FGD, 18th December, 2023).

5. Discussion

5.1 Perceived usefulness of ChatGPT for completing assessment tasks

The use of the ChatGPT tool in completing assessment tasks is portrayed as an opportunity and a threat as well (Tarisayi, 2024). This suggests that it is possible that some student teachers are not aware of the consequences of using ChatGPT in relation to the university’s academic integrity guidelines. On the other side, they could be aware of the guidelines but they fail to comply with them due to various reasons including the need to use shortcuts in

Table 4. ChatGPT Usage in completing assessment tasks is against academic integrity

Responses	Counts	%	Valid percent	Cumulative percent
Yes	174	45.4	45.7	45.7
No	192	50.1	50.4	96.1
Don’t know	15	3.9	3.9	100.0
Total	381	99.5	100.0	
Missing system	2	0.5	0.5	
Total	383	100.0		

Source: Authors’ own work

completing the assignments, pressure to get good grades and inability to locate credible sources.

During the FGDs, some participants defended themselves comfortably that being able to find AI tools and use them, is also a sense of creativity. This view partly reflects the constructivism theory which implies that students are free to interact with various resources to make meaningful construction of knowledge. Student teachers believe that the incoming of AI tools especially ChatGPT is an opportunity that should not be wasted because it simplifies their work. This finding is consistent with findings from some previous studies. For example, one study explored perceptions and intentions to use ChatGPT whereby participants said that the AI tool makes their work easier, and faster and does not distress them (Bonsu and Baffour-Koduah, 2023). Also, another empirical study surveyed 49 undergraduate students from Massachusetts University to obtain students' opinions and perspectives on the use of ChatGPT. The results showed that 40 (81.6%) of respondents suggested that ChatGPT is useful and therefore instructors should allow students to use it for written assessment tasks (Kanabar, 2023).

Moreover, a qualitative descriptive study investigated English Education students' Perception of the Use of ChatGPT in Writing Articles. The findings indicated that 60% of the students supported that ChatGPT simplifies writing scientific papers and 40% of respondents appreciated that texts generated by the AI tool are easy to grasp (Hasanah and Nurcholis, 2024). Observations from the preceding studies suggest that the majority of students appreciate the great potential of ChatGPT usage in their academic endeavours. Despite its usefulness, its ethical use is indispensable to safeguard institutional academic integrity. The AI tool should be used for generation of ideas and ought to be acknowledged whenever necessary.

The other percentage of the students who do not use ChatGPT may suggest that they are aware of academic integrity and honesty in their accomplishment of assigned tasks. On the other hand, some student teachers knew the AI tool but did not like to use it due to its shortcomings. Others had no idea about the existence of the tool or had limited devices to access the AI tool. This implies that awareness of academic integrity, knowledge of ChatGPT use and the existence of supportive devices that enhance access to the AI tool could affect an individual's perception of using ChatGPT. Altogether, it is important for all, regardless of their perceptions to abide by academic rules to maintain academic integrity while using the AI tool.

5.2 Using ChatGPT to complete assessment tasks and the consideration of academic integrity

Students extract information from the ChatGPT by copying and pasting it for final submission. Their work lacks citations, and factual references and sometimes presents inaccurate information. This practice is against academic integrity which requires an individual to cite and locate sources used to prepare any piece of academic work. The foregoing finding resonates with the findings from the study which investigated how university students perceive using chatGPT for learning purposes. Ngo (2023) reported that ChatGPT is unable to produce real references, cannot cite sources and sometimes does not provide references at all.

Further, even though the AI tool has a wonderful ability to generate answers to various assessment tasks it does not help the student to be autonomous. It lowers students' ability to think and work independently and reflects on the content they learn without overdependence on outsourcing. Regarding constructivism theory which advocates for active, collaborative and responsible learning, the use of ChatGPT as a shortcut to complete academic tasks

underestimates the active role of students in communicating, cooperating and using resources appropriately to construct knowledge. Overdependence on using ChatGPT limits students' personal development and renders students unable to help one another harness constructive ideas required to address their queries. This finding is in line with the finding from the study by [Hasanein and Sobaih \(2023\)](#) which used a critical literature review to explore the importance of ChatGPT in academic writing, its effects on human creativity and academic integrity. The findings revealed that the misuse of AI tools kills critical and creative academic writing, as well as independent thinking.

Additionally, the tool is condemned for doing partial work, because it does not provide a source of information to be used to support and justify students' work ([Xu et al., 2024](#)). Participants reported that sometimes incorrect information and some examples provided by ChatGPT are out of students' context hence it is thought to have a possibility of misleading. Consequently, misleading information jeopardizes academic integrity and students' knowledge acquisition. As mentioned elsewhere, ChatGPT is not complete in itself, it needs human input to function. As a result, its existence in educational settings should be considered to enhance academic integrity.

5.3 Study limitations

Given the sensitivity of the topic on academic integrity some participants were hesitant to share information until when the researchers clarified the purpose of the study. Participants were told that the information they would provide was purely for academic purpose. Also, this study was conducted when students' regular classes were in progress, such that it was challenging to set universal time to conduct FGDs where all intended participants could attend. The researcher was flexible enough to find hours which were conducive for participants to participate in FGDs.

5.4 Implications for research, practice and society

Future research should include other universities in Tanzania to obtain a broader understanding of the phenomenon. Also, other data collection techniques such as observations, documentary review, in depth interviews should be incorporated in future studies to enable more rigorous data triangulation. Further, the subsequent studies may incorporate other disciplines and levels of study to obtain a more balanced representation.

For practitioners, given the fact that ChatGPT is a relatively new AI tool, training for raising awareness of its use should be conducted. Besides, specific academic integrity guidelines for its ethical use should be formulated and implemented. Likewise, instructors should set questions that prompt students original thinking which the AI tool cannot easily give contextualized solutions. The university where the AI tool is used is an open system that is within the society, and the impacts of AI technologies are inevitable in social domains. The society, therefore, should be informed about the emerging AI tools with their associated potentials and shortcomings to safeguard the norms and interests of the society.

5.5 Contribution to the body of knowledge

The study provides a context-specific overview of student teachers' perceptions of ChatGPT use and its implication on academic integrity. Further, it provides valuable insights for developing and reviewing academic integrity policies and guidelines for using AI tools including ChatGPT in education. Further, the study findings would help to encourage discussion on the responsible and ethical use of ChatGPT in various academic endeavours.

5.6 Conclusions and recommendations

This study investigated academic integrity considerations of ChatGPT usage in assessment activities among the finalist student teachers at the University of Dar es Salaam. The findings indicate that students had varied opinions on the use of ChatGPT as far as academic integrity is concerned. Although, ChatGPT is a popular and useful tool among student teachers, it should not replace students' responsibility of searching and presenting genuine knowledge. They need to interrogate and put a critical voice to any piece of information they come across. With this regard, ChatGPT should not be used as a standalone tool to accomplish assessment tasks, rather students should be guided to use it for generation of ideas. Given the fast growing field of AI in education, intervention should take place accordingly, to avoid the possibility of overreliance on ChatGPT use among student teachers. In so doing it will enable to safeguard students' ability to think critically, learn independently and construct knowledge.

Also, based on the findings, the researcher recommends that: Challenges related to the use of AI tools call for joint efforts and it should not be confined to an individual, community, organization or a particular nation alone, because the AI technology is ubiquitous. For example, being aware of its challenges, Audrey Azoulay, UNESCO'S Director General gave one of her recommendations about AI as follows:

The world needs stronger ethical rules for Artificial Intelligence: this is the challenge of our time. UNESCO's recommendation on the ethics of AI sets the appropriate normative framework. Our member states all endorsed this recommendation in November 2021. It is high time to implement the strategies and regulations at the national level. We have to walk the talk and ensure we deliver on recommendation's objectives (UNESCO, 2022, p. 5).

Thus, UNESCO calls for joint efforts across nations to implement rules guiding the ethical use of AI tools for the betterment of quality education. In specific terms, the following recommendations could help to minimize academic misconduct brought about by the irresponsible and unethical use of ChatGPT. Students should be aware of ChatGPT limitations to enable them to use it ethically and responsibly. They should understand that ChatGPT is just an aid that cannot replace their natural intelligence. They are supposed to evaluate critically the responses produced by the ChatGPT and judge their suitability for the tasks at hand.

Likewise, the university should support in raising awareness of ChatGPT and other emerging AI tools to students and instructors. It should encourage instructors to provide assessment tasks that require students' critical and inquiry mind by providing questions that are context specific, unlike general questions that can be easily solved by the AI tool. Further, the university should continuously review its academic integrity guidelines to address ChatGPT usage and other incoming AI tools.

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